Three-Axis Model of m6A Epitranscriptomic Regulation in mCRPC

Three writer-weighting approaches compared — best model selected by cross-validated AUC

A. Model 3 — Logistic Regression Writer Weights (selected model)

Train L2-regularized logistic regression on 7 writer z-scores to predict histology:

$$\log\left(\frac{P(\mathrm{SCNC})}{1-P(\mathrm{SCNC})}\right)=\beta_0+\beta_1\,z_{\mathrm{METTL3}}+\beta_2\,z_{\mathrm{METTL14}}+\cdots+\beta_7\,z_{\mathrm{CBLL1}}$$

Convert coefficients to writer weights (absolute value, normalized):

$$w_i=\frac{|\beta_i|}{\sum_{j=1}^{7}|\beta_j|}\times\sum_{j=1}^{7}w_j^{\mathrm{manual}}$$ (CV-AUC = 0.710 ± 0.120, 5-fold, $n=538$)

RBM15	CBLL1	ZC3H13	WTAP	METTL3	METTL14	RBM15B
β = +0.652	β = -0.619	β = -0.299	β = +0.286	β = +0.245	β = +0.204	β = -0.037
w = 2.73	**w = 2.59**	**w = 1.25**	**w = 1.19**	**w = 1.03**	**w = 0.85**	**w = 0.16**
(Targeting)	(E3 Ligase)	(Scaffold)	(Scaffold)	(Catalytic)	(Allosteric)	(Targeting)

A2. Model 1 — Manual Biology-Based Weights (comparison)

Fixed weights assigned from known biochemistry of the writer complex:

METTL3	METTL14	WTAP	ZC3H13	RBM15	RBM15B	CBLL1
w = 3.0	w = 2.0	w = 1.5	w = 1.0	w = 1.0	w = 0.8	w = 0.5
(Catalytic)	(Allosteric)	(Scaffold)	(Scaffold)	(Targeting)	(Targeting)	(E3 Ligase)

No learning — weights reflect textbook subunit importance (HeLa-derived, not mCRPC-specific)

A3. Model 2 — Bottleneck (Complex Assembly) (comparison)

Step 1 — Convert each writer z-score to an "activity probability" via the normal CDF:

$$P_i=\Phi(z_i)\qquad (z=0\Rightarrow P=0.5;\quad z=+2\Rightarrow P\approx 0.98;\quad z=-2\Rightarrow P\approx 0.02)$$

Step 2 — Weighted geometric mean (bottleneck-sensitive: any low subunit drags down the whole complex):

$$\text{Complex Assembly}=e^{\frac{\sum_{i=1}^{7}w_i\cdot\ln(P_i)}{\sum_{i=1}^{7}w_i}}\qquad (w_i=\text{manual weights})$$

Step 3 — Eraser probability and Net Deposition:

$$\text{Eraser Assembly}=\text{mean}(\Phi(z_{\mathrm{FTO}}),\ \Phi(z_{\mathrm{ALKBH5}}))\qquad \text{Net Dep}_{\text{bottleneck}}=\text{Complex Assembly}-\text{Eraser Assembly}$$

Uses manual weights but with a nonlinear functional form — weakest-link sensitivity, compressed [0, 1] range

B. Axis 1 — Net m6A Deposition (using LR weights)

$$\text{Net Deposition}=\frac{\sum_{i=1}^{7}w_i\cdot z_i}{\sum_{i=1}^{7}w_i}-\bar{z}_{\text{Erasers}}$$

$z_i=\frac{x_i-\bar{x}_i}{\sigma_i}$ (per-gene z-score across 634 patients) $\quad\bar{z}_{\text{Erasers}}=\text{mean}(z_{\mathrm{FTO}},\ z_{\mathrm{ALKBH5}})$

C. Axis 2 — Oncogenic Readout

$$\text{Oncogenic Readout}=\bar{z}_{\text{Onc. Readers}}-\bar{z}_{\text{Supp. Readers}}$$

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Oncogenic:     IGF2BP1,  IGF2BP2,  IGF2BP3,  YTHDF1

Suppressive:   YTHDF2,   YTHDF3,   YTHDC1,   YTHDC2
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D. Axis Weight Optimization + Axis 3 — Functional Impact

Train a second logistic regression on the two standardized composite axes:

$$\log\left(\frac{P(\mathrm{SCNC})}{1-P(\mathrm{SCNC})}\right)=\alpha_0+\alpha_1\cdot\widetilde{\text{NetDep}}+\alpha_2\cdot\widetilde{\text{OncRead}}\qquad \tilde{x}=\text{StandardScaler}(x)$$ CV-AUC $=0.576\pm0.045$

$$W_{\mathrm{ND}}=\frac{|\alpha_1|}{|\alpha_1|+|\alpha_2|}=0.435\qquad W_{\mathrm{OR}}=\frac{|\alpha_2|}{|\alpha_1|+|\alpha_2|}=0.565$$

$$\boxed{\text{Functional Impact}=0.435\times\text{Net Deposition}+0.565\times\text{Oncogenic Readout}}$$

E. Liver Effect — Independent of Subtype (OLS Regression)

Test whether liver independently predicts Oncogenic Readout after adjusting for confounders:

$$\text{Oncogenic Readout}=\beta_0+\beta_1\cdot 1_{\text{liver}}+\beta_2\cdot 1_{\text{SCNC}}+\beta_3\cdot 1_{\text{Basal}}$$

Results (n = 538):

Variable	Coef	SE	p	
is_liver	**+0.370**	**0.101**	**2.62e-04**	*******
is_SCNC	+0.114	0.113	3.11e-01	ns
is_Basal	+0.074	0.070	2.94e-01	ns

→ Liver is the only significant predictor — effect is independent of histology and subtype

F. RBM15 vs RBM15B — Paralog Coordination (Fisher's Exact Test)

Classify each patient as high (top quartile, z ≥ 0.60) or low for each paralog:

	RBM15B high	RBM15B low
RBM15 high	**42**	**106**
RBM15 low	127	359

Odds Ratio and Fisher's exact p-value:

$$\mathrm{OR}=\frac{a\times d}{b\times c}=\frac{42\times 359}{106\times 127}=1.120\qquad p_{\text{Fisher}}=0.598\ (\text{ns})$$

→ OR ≈ 1, p = ns: RBM15 and RBM15B are expressed INDEPENDENTLY (not coordinated, not mutually exclusive)

Dataset: 634 mCRPC patients | 479 Adenocarcinoma, 59 SCNC | Expression: log2 CPM, z-scored | L2 regularization (C = 1.0)